Python Course for Physicists (And everyone else!)

Chunk 12: Scipy & Regressions

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Matplotlib: rcParams

- rcParams: Set style of plots for complete document:

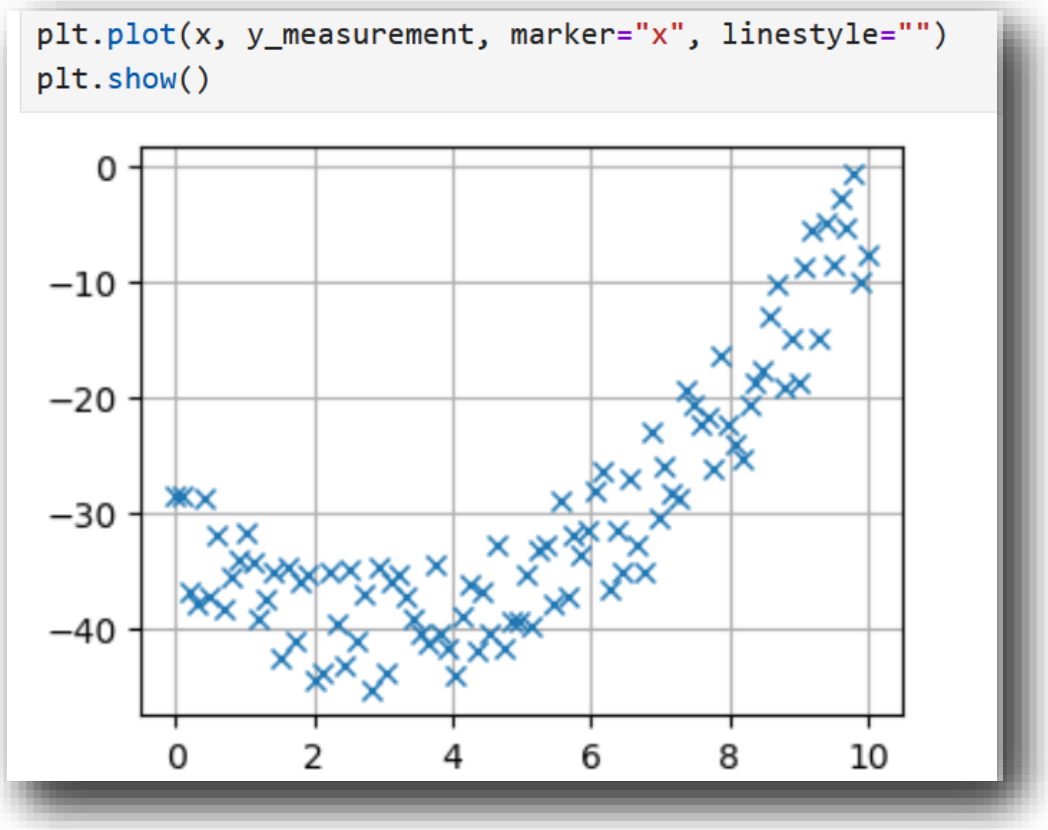
```
%matplotlib widget
import matplotlib.pyplot as plt
plt.rcParams["figure.figsize"] = (4,3)
plt.rcParams["axes.grid"] = True
```

- Complete set of parameters here.

Live Coding: Visualizing Noisy Data

- *plt.plot(x, y, marker="x", linestyle="")*

```
plt.plot(x, y_measurement, marker="x", linestyle="")
plt.show()
```



Finding the Function: Regression

- Usually: We have an idea of the “nature” of our data.
  → Example shown: “looks like a quadratic function”
- But: We don’t know the constants describing the polynomial
- Two options:

1. np.polyfit(x, y, order) → returns the coefficients
2. General function definition → scipy.optimize.curve_fit(…)

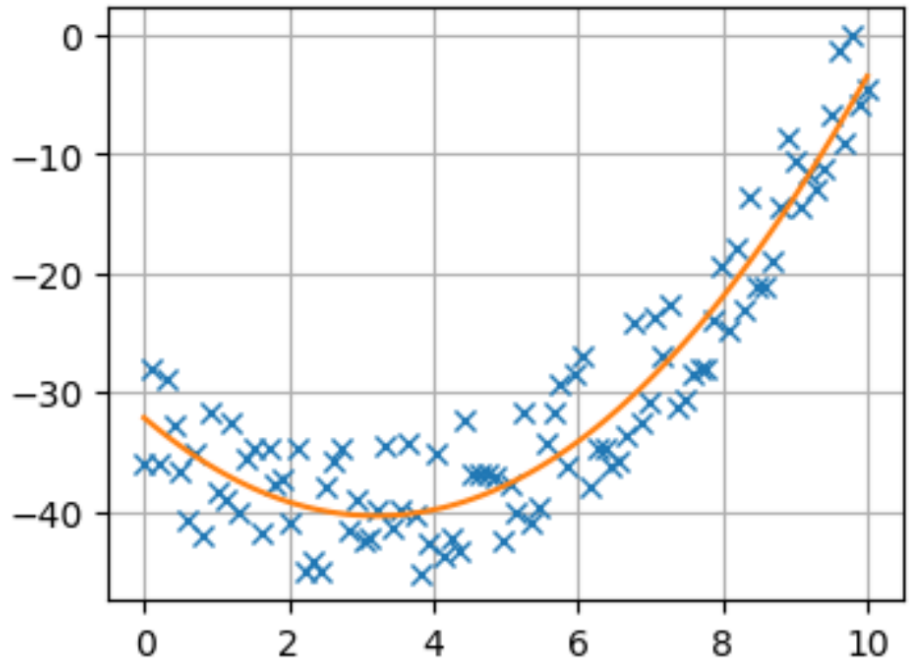
Live Coding: Numpy polyfit

- polyfit finds coefficients.
- Coefficients can be used to create a fitted function
- Plot the fitted function

```
coeffs = np.polyfit(x, y_measurement, 2)
print(coeffs)
y_fit = coeffs[0] * x**2 + coeffs[1] * x + coeffs[2]

plt.plot(x, y_measurement, marker="x", linestyle="")
plt.plot(x, y_fit)
# plt.plot(x, y_theory);
```

```
[  0.80085884  -5.14075265 -32.11890331]
[<matplotlib.lines.Line2D at 0x1a30dd41090>]
```



scipy (constants)

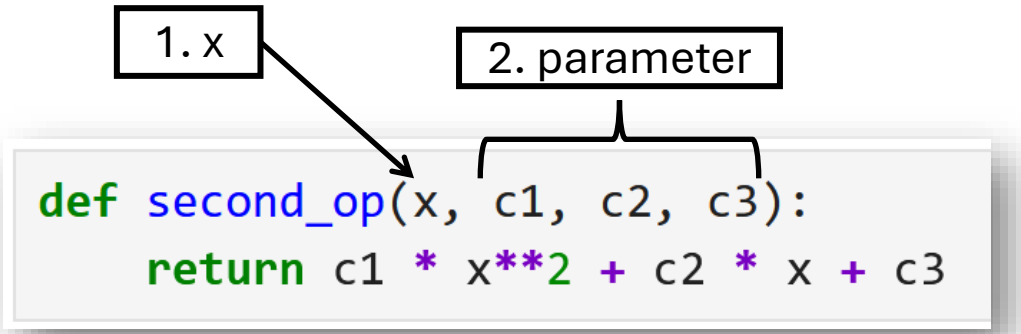
- scipy → scientific python
- Tons of useful functions
- One example: scipy.constants:

```
import scipy.constants as c
print(c.c)
```

```
299792458.0
```

Live Coding: scipy curve_fit

- Need to create a python function, that describes our mathematical function
  → Order matters
- Create coefficients & covariance
  - Covariance can be used to calculate confidence intervals
- If convergence fails: Give initial guess with optional p0 parameter



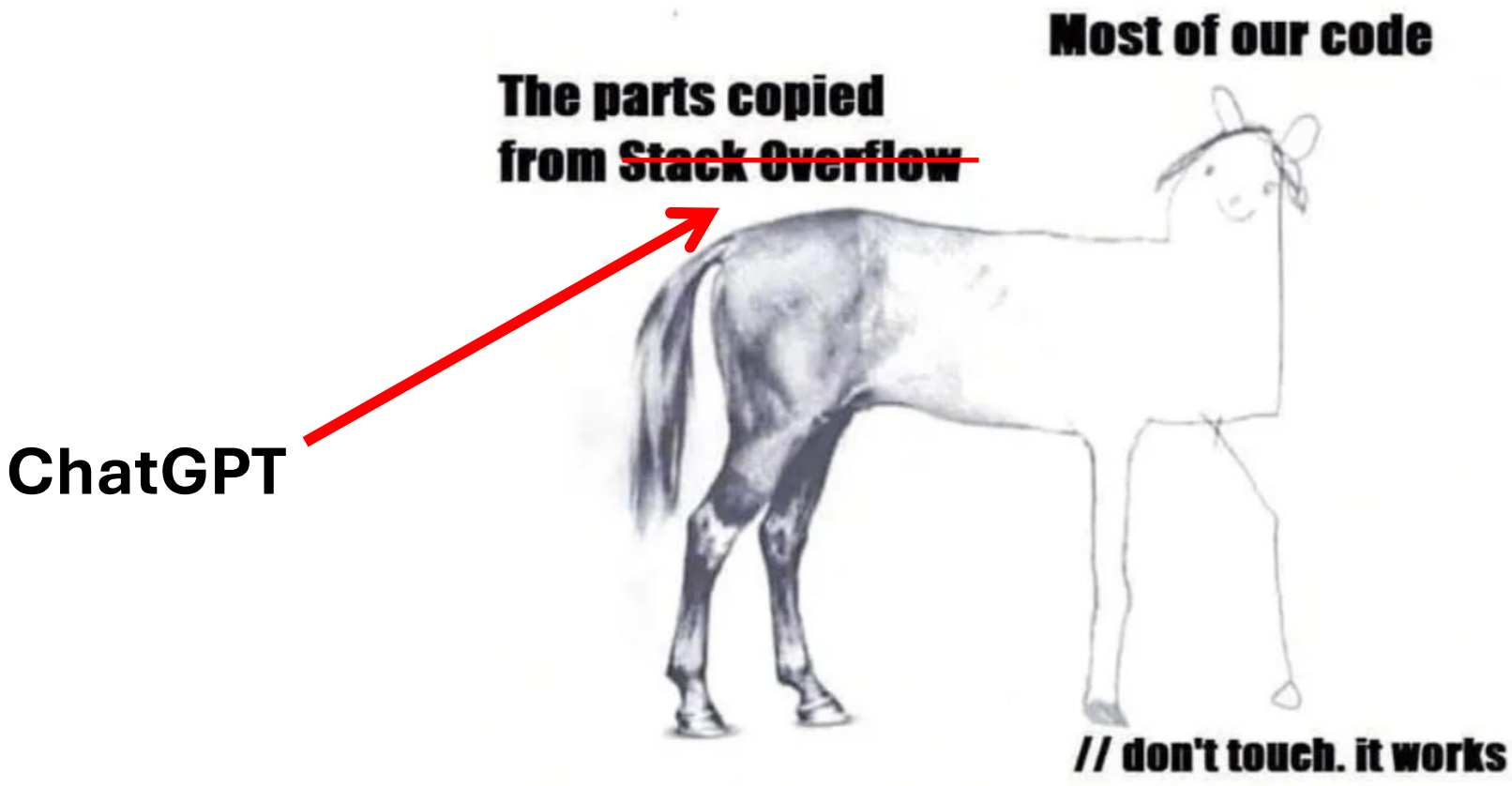

```
import scipy

def second_op(x, c1, c2, c3):
    return c1 * x**2 + c2 * x + c3

coeffs, cv = scipy.optimize.curve_fit(second_op, x, y_measurement)
print(coeffs)
```

```
[  0.80085884  -5.14075261 -32.11890338]
```

Good Luck with your Tasks ☺



[https://python.plainenglish.io/20-funny-programming-memes-that-will-make-your-day-5793a8e1608b]